STA201: Elements of Statistics and Probability
Lecture 7

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Correlation Analysis

Why correlation?

There are many situations where the objective of the study is to see the relationship among variables.

Consider the following situations-

- Is there any relation between shoe size and height?
- Is there any relation between number of hours you exercise every day and your heart rate?
- Is there any relation between number of hours you study and your grade?
- Is there any relation between math scores and overall scores in exam?
- Is there any relation between temperature in summer and water consumption?

Correlation Analysis

- The statistical tool for discovering and measuring the **relationship between two or more variables** is studied is called correlation analysis.
- Correlation shows whether and how strongly pairs of variables are related.
- It is concerned with **two (or more) sets of data**, not with only one set.
- When we measure inter-relationship between
  - two variables, then it is called **simple correlation**
  - among **more than two variables** is termed as **multiple correlation.**

Advantages of correlation analysis

- Correlation analysis measures the strength of relationship between the variables.
- It also provides the direction of the relationship between the variables.
- It reduces the range of uncertainty in matter of prediction. Prediction is not perfectly accurate here, but helpful.

Disadvantage of correlation analysis

- Correlation only describes a relationship. It does not tell us why.
- Correlation is affected greatly by the range of the data set.
- If there is outlier in the data set, it significantly alters the correlation.
- Correlation does not capture strong non-linear relationship between variables.

Types of Correlation Analysis

1. Positive correlation – Variables change in the same direction

- As X is increasing, Y is also increasing
- As X is decreasing, Y is also decreasing

Examples

- Income and expenditure on luxury goods
- A students study hour and his test score
- Advertising and sales
- Water consumption and temperature

Types of Correlation Analysis Cont...

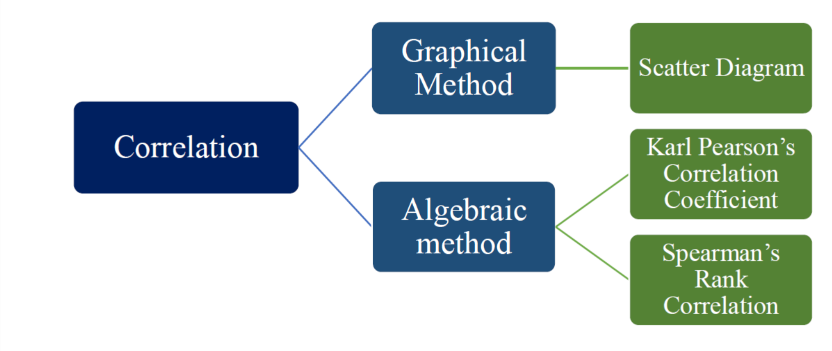
2. Negative correlation – Variables change in the opposite direction

- As X is increasing, Y is decreasing
- As X is decreasing, Y is increasing

Examples

- Price and demand of a commodity
- Students absence in class and their grades
- TV registration and cinema attendance
- Alcohol consumption and driving ability

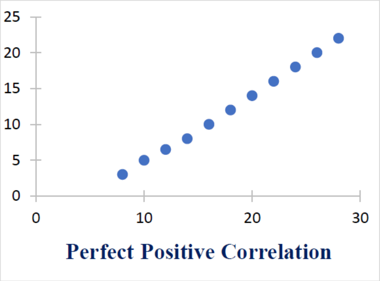
How to determine correlation

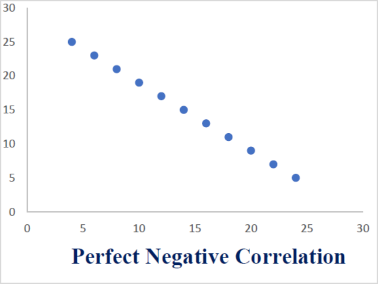


Scatter diagram method

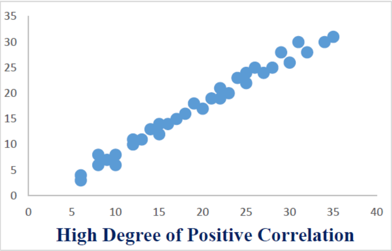
- It is the simplest way for ascertaining if the two variables are related or not.
- The given data are plotted on a graph paper in the form of dots.
- The greater the scatter of points over the graph, the lesser is the relationship between the variables.

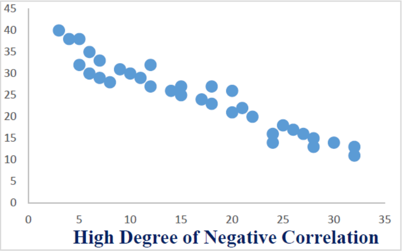
Scatter diagram method Cont...



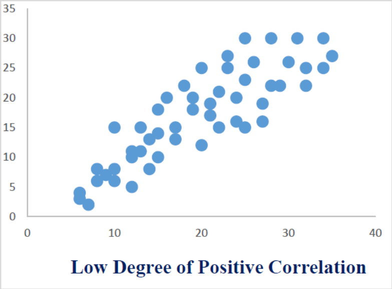


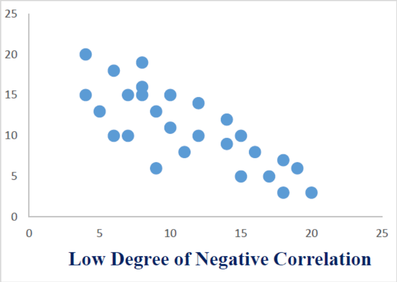
Scatter diagram method Cont...



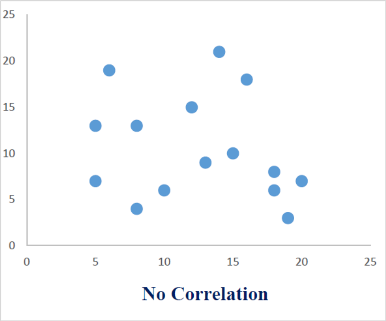


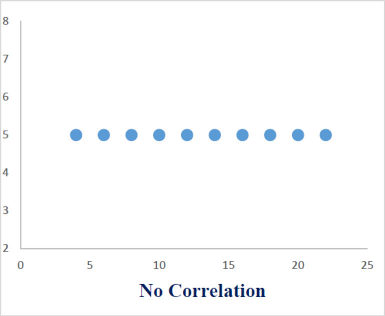
Scatter diagram method Cont...





Scatter diagram method Cont...





Scatter diagram method Cont...

Merits

- Simple and non-mathematical method.
- Easily understandable.
- Not influenced by extreme values.

Demerits

- Can't establish the exact degree of correlation between the variables

Example 1

Let X and Y represent the two variables. Examine the relationship between these two variables by plotting a scatter diagram.

Income ('000' Tk.) X	Expenditure ('000' Tk.) Y
5	4
10	8
15	12
20	18
30	20
50	45

Example 1 Cont...

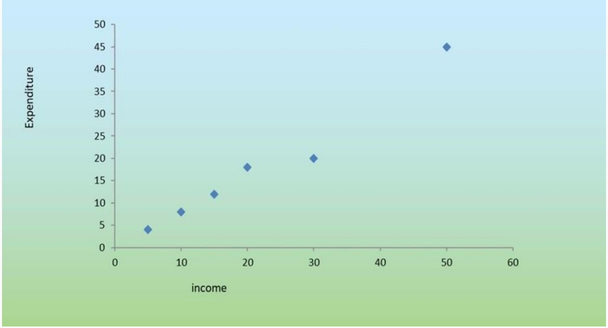


Figure 1: Scatter diagram between income and expenditure

- The scatter diagram shows that there exists a positive correlation between income, x and the expenditure, y.
- Which means for higher income, expenditure will also be higher or vice−versa.

Example 2

The price and amount of demand is given below. Draw a scatter diagram and comment:

Price ($)	Demand (Kg)
20	4
19	4
18	5
16	6
14	6
12	8
10	9
10	9
9	10
8	10

Example 2 Cont...

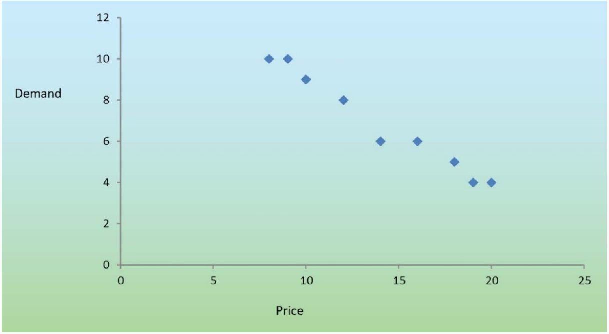


Figure 2: Scatter diagram between price and demand

In this case, the correlation is negative. There is negative relationship between the price, x and the amount demand, y, when price increase, demand decreases.

Karl Pearson's Correlation Coefficient

- It is also known as Pearson's product-moment correlation coefficient.
- It was introduced by Galton in 1877 and developed later by Karl Pearson in 1895.
- This is the most popular measure of correlation for measuring the degree or strength of linear relationship between two variables.

Assumptions of Pearson's Correlation Coefficient

- The underlying relationship between two variables is linear, which is the assumption of linearity.
- The variables under consideration should have a bivariate normal distribution.

The Pearson's correlation coefficient is denoted by r,

$$r = \frac{\sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y}}{\sqrt{(\sum_{i=1}^{n} x_i^2 - n\bar{x}^2)(\sum_{i=1}^{n} y_i^2 - n\bar{y}^2)}}$$

Interpretation of r

Degrees	Positive	Negative
Absence of correlation	0	0
Perfect Correlation	+1	-1
High Degree	+0.75 to +1	-0.75 to -1
Moderate Degree	+0.25 to +0.75	-0.25 to -0.75
Low degree	0 to +0.25	0 to -0.25

So, the values of r lies between -1 to 1. That means $-1 \leq r \leq 1$

Merits and Demerits of r

Merits

- It summarizes in one value the degree of correlation and direction of correlation also.
- It gives us exact measure of degree of association.
- It is independent of units of the variables.

Demerits

- Value of r is affected by extreme values.
- Calculation method is long and time consuming.
- Only applicable for **quantitative measurements**.
- When the population is **not normally distributed,** r may underestimate the relationship between the variables

Coefficient of Determination (r^2)

- **The coefficient of determination is the square of the correlation coefficient r , that is r^2 .**
- It is the measure of the strength of the relationship between two variables.
- It measures the percentage of variation of dependent variable which can be explained by independent variable.
- The value of coefficient of determination lies between 0 and 1.
- Suppose $r = 0.9$, then $r^2 = 0.81$. This would mean that 81% of total variation in y can be explained by x.

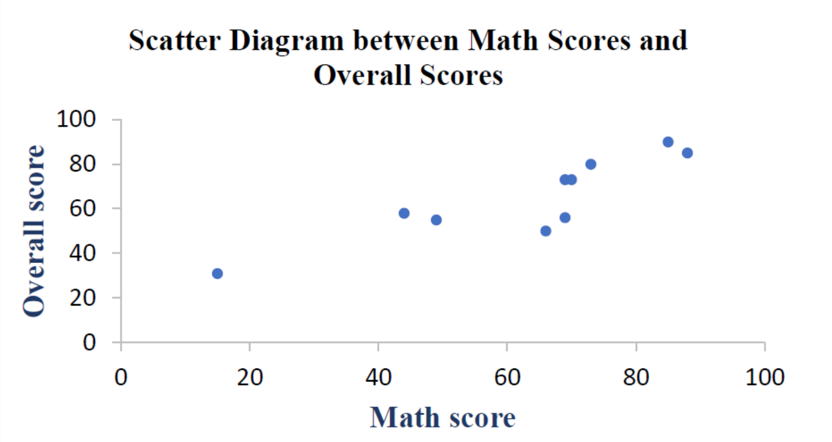
Example

A small experiment is conducted involving 10 students to investigate the association between students math scores and overall scores in exam.

Math Scores (%)	70	85	66	15	69	49	73	44	88	69
Overall scores (%)	73	90	50	31	56	55	80	58	85	73

Our aim is to find out whether there is any linear association between math score and overall score.

Solution: Scatter plot



Comment: This graph shows positive correlation between math scores and overall scores.

Solution

x	y	xy	x^2	y^2
70	73	5110	4900	5329
85	90	7650	7225	8100
66	50	3300	4356	2500
15	31	465	225	961
69	56	3864	4761	3136
49	55	2695	2401	3025
73	80	5840	5329	6400
44	58	2552	1936	3364
88	85	7480	7744	7225
69	73	5037	4761	5329
$\sum x = 628$	$\sum y = 651$	$\sum xy = 43993$	$\sum x^2 = 43638$	$\sum y^2 = 45369$

Solution

Here,
$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{628}{10} = 62.8$
$\bar{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{651}{10} = 65.1$

$$r = \frac{\sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y}}{\sqrt{(\sum_{i=1}^{n} x_i^2 - n\bar{x}^2)(\sum_{i=1}^{n} y_i^2 - n\bar{y}^2)}}$$
$$r = 0.878$$
$$r^2 = 0.771$$

Interpretation

Correlation coefficient (r)

There exists high degree of positive correlation between math scores (x) and overall scores (y) of students.

Coefficient of determination (r^2)

77.1% of total variation in overall score (y) can be explained by the linear relationship between math scores (x) and overall scores (y). The other 22.9% of the total variation in overall score (y) remains unexplained.

Thank You